DL836 Y4 – Research & Analytics
CA1 – Data Analytics for Business

Assessment Overview

Modern organisations increasingly use data analytics, machine learning, recommendation systems and AI-assisted development tools to support strategic decision-making.
In this assessment, you will assume the role of a Data Analytics Consultant working for a retail organisation.
Using Python, Pandas, Scikit-Learn and GitHub Copilot, you will:

- Clean and prepare business datasets
- Perform exploratory data analysis
- Develop predictive models
- Build recommendation systems
- Generate business insights
- Develop executive-level recommendations
- Critically evaluate your use of AI tools

Learning Outcomes

4. Apply analytical techniques and software tools to solve analytical problems.
5. Analyse business problems by applying analytical methods to find solutions.

Dataset Allocation

Each student will receive 43 files to use to complete this assignment. Students must only use the datasets allocated to them.:

1. **Retail Performance (3 Datasets)**
   - Retail_outlets.csv; Retail_regions.csv & Retail_stores.csv

2. **Recommendation System (3 Datasets)**
   - Customer Dataset; Product Dataset & Transaction Dataset

3. **Business Scenario (1 executive**
   - Marketing Director (or), Commercial Director (or), Operations Director

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Marketing Component	Weight
Task 1 – Retail Performance Analytics	25%
Task 2 – Recommendation System Development	25%
Task 3 – Business Recommendations	30%
Task 4 – AI Process Log, GitHub Repository & Reflection	20%
CA1 Total	**100%**

<u>Submission:</u>

<u>Github</u>*: TASK 1 & 2 clearly identified*

Hub (Turnitin) *– TASK 3 & 4 - only submit 1 report document.*

<u>Due Date</u>: Sun 15th November, 2025 (Midnight)

CA Percentage for the Module: 50%

Use of Generative AI	AI can **ONLY** be used for **TASK 1 & 2 - Code Development and AI Prompts.** **No** AI use for **TASK 3 & 4 – Business Recommendations & AI Process Log and Reflection.** Generative AI use in this assessment for TASK 3 & 4 is considered a breach of the IADT Academic Integrity Policy.

Task 1 – Retail Performance Analytics (25%)

Identify the factors that drive retail business performance and develop predictive models capable of forecasting future outcomes.

Section	Requirements	Suggested Output
Data Preparation & Quality Assessment	Load and inspect the dataset, analyse data quality, handle missing values, duplicates and outliers, and create at least 3 engineered features.	Missing value analysis, duplicate analysis, outlier investigation, data quality summary and feature engineering.
Exploratory Data Analysis	Investigate regional, revenue, seasonal, store and marketing performance. Complete 4 GroupBy analyses and identify key business trends.	Summary statistics, GroupBy outputs, Top 5 stores/regions, and interpretation of findings.
Pivot Tables & Correlation Analysis	Create at least 2 Pivot Tables and investigate relationships between variables.	Pivot Tables, correlation matrix/heatmap and discussion of variables most strongly associated with Sales Revenue.
Predictive Analytics	Develop and compare at least 1 predictive models (e.g. Linear Regression, Decision Tree, Random Forest).	Model selection rationale, train/test methodology, (could use MAE, RMSE, R^2), and interpretation of results.
Feature Importance & Business Insights	Identify key drivers of performance that is important for decision making & business significance.	Feature importance analysis (Sales by marketing spend, regions, etc.) for potential management actions. (Write up of actions in TASK 3 ONLY)
Visualisation	Create a minimum of 3 meaningful visualisations.	Visualisations supporting business performance and predictive analytics findings. All charts must be labelled and interpreted.

Task 2 – Recommendation System Development (25%)

Develop and evaluate recommendation systems to support customer engagement and increase business value that can be used to report back to business managers.

Section	Requirements	Suggested Output
Data Preparation & Integration	Load, inspect and combine the customer, product and transaction datasets. Create a customer-product interaction matrix and assess data quality.	Data quality assessment, dataset integration process and interaction matrix.
Customer Analytics	Analyse customer purchasing behaviour and identify high-value customers.	Top 10 customers by spend, Top 10 customers by purchase frequency and customer value analysis.
Product Analytics	Analyse product performance and identify key sales trends.	Top 10 products by revenue, revenue by category and revenue by brand.
GroupBy & Pivot Table Analysis	Complete minimum of 3 GroupBy analyses and **2** Pivot Tables**.**	GroupBy outputs and Pivot Tables with interpretation of findings.
Recommendation System Development	Develop 1 popularity-based recommendation systems (Collaborative or Content filtering) and one additional recommendation algorithm.	Working recommendation systems and explanation of why this recommendation system was used.
Recommendation Evaluation	Compare recommendation approaches and justify which method performs best.	Recommendations for at least 5 example customers, evaluation and comparison of approaches.
Business Interpretation	Explain customer behaviour patterns and the business value of recommendations.	IN TASK 3 ONLY (Discuss strengths, limitations and business usefulness).

Task 3 – Business Recommendations (30%)

Based on the business officer assigned to you, produce a 1,000 – 1,500-word report to interpret the Task 1 & 2 findings into recommendations for the business executive. More details will be provided in class.

Section	Requirements	Suggested Output
Executive Summary	Focus on the assigned stakeholder (Marketing Director, Commercial Director or Operations Director).	Demonstration of understanding of stakeholder priorities and business objectives.
Evidence-Based Recommendations	Develop a minimum of **5 recommendations** supported by analytical evidence.	Five recommendations linked to analysis results.
Use of Retail Analytics Findings	Incorporate findings from Task 1.	References to key revenue drivers, trends and model outputs.
Use of Recommendation System Findings	Incorporate findings from Task 2.	References to customer behaviour, recommendation outputs and product insights.
Business Justification	Explain the rationale for each recommendation.	Expected business impact, benefits and risks.

Section	Requirements	Suggested Output
Summary	Present recommendations clearly and professionally.	Executive-summary style communication suitable for senior management.

Task 4 – AI Process Log & Reflection (10%)

To demonstrate responsible use of AI tools and software development practices.

Section	Requirements	Suggested Output
AI Process Log	Record significant use of AI tools during the project.	AI Process Log completed using the provided template.
Validation & Debugging	Demonstrate how AI-generated outputs were tested, verified and improved.	Examples of validation, debugging and corrections made.
GitHub Repository	Maintain a GitHub repository throughout the project.	Repository link, organised project structure and development history.
Version Control	Demonstrate evidence of incremental development.	Minimum **3 meaningful commits** with appropriate commit messages.
Reflection	Critically evaluate the use of AI in the project.	Reflection addressing productivity, limitations, validation and professional judgement.

Any research materials and sources available to support the information in your report.
All materials used must be properly cited and referenced using APA style. For style details refer to:
\\Sideshowbob1\lecturers\Tim McNichols\CC4\CA Referencing Information

Marking Rubric

Criteria	A (80-100%)	B+ (70-79%)	B (60-69%)	B- (55-59%)	C+ (50-54%)	C (40-49%)	D (35-39%)	F (0-34%)
Task 1: Retail Performance Analytics (25%)	Excellent analytical workflow with excellent implementation of data preparation, modelling and evaluation. Excellent interpretation of business implications.	Very strong analytical workflow with excellent implementation of data preparation, modelling and evaluation. Strong interpretation of business implications.	Good overall implementation with most required outputs completed correctly. Predictive modelling and interpretation are generally effective.	Competent implementation with most task requirements completed. Some weaknesses in modelling, evaluation or interpretation.	Basic implementation meeting minimum requirements. Limited depth in analytics and interpretation.	Limited implementation with several missing outputs or weak analysis.	Minimal evidence of analytical work completed. Significant omissions.	No meaningful evidence of required analytical work.
Task 2: Recommendation System Development (25%)	Excellent recommendation system implementation with excellent evaluation and comparison of techniques. Excellent business insight demonstrated.	Strong recommendation system implementation with effective evaluation and comparison of techniques. Good business insight demonstrated.	Good implementation of recommendation methods with reasonable evaluation and interpretation.	Competent implementation but limited evaluation or comparison of approaches.	Basic recommendation system implemented meeting minimum requirements.	Limited implementation and incomplete evaluation.	Minimal evidence of recommendation system development.	No meaningful recommendation system demonstrated.
Task 3: Executive Business Recommendations (30%)	Excellent recommendations supported by analytical evidence and clearly linked to stakeholder requirements. Excellent business understanding demonstrated.	Very strong recommendations supported by analytical evidence and clearly linked to stakeholder requirements.	Good recommendations with clear links to analytical findings and business context.	Competent recommendations but some lack depth, justification or business relevance.	Basic recommendations meeting minimum requirements but lacking strong supporting evidence.	Limited recommendations with weak links to analysis.	Recommendations poorly justified or incomplete.	No meaningful recommendations provided.
Task 4: AI Process Log, GitHub Repository & Reflection (20%)	Excellent AI documentation, repository management and	Strong AI documentation, repository management and	Good AI Process Log and repository structure. Reflection	Competent documentation and repository evidence with some	Basic documentation meeting minimum requirements	Limited documentation and weak evidence of	Minimal process log, repository activity or reflection evident.	No meaningful submission.

Marking Rubric

Criteria	A (80-100%)	B+ (70-79%)	B (60-69%)	B- (55-59%)	C+ (50-54%)	C (40-49%)	D (35-39%)	F (0-34%)
	reflection with clear evidence of validation and development practice. Excellent professional judgement demonstrated.	reflection with clear evidence of validation and development practice.	demonstrates understanding of AI-assisted development.	limitations in reflection or validation.	including GitHub activity and reflection.	development process.		